CLOUD & DEVOPS

Centralized Logging Visual Guide

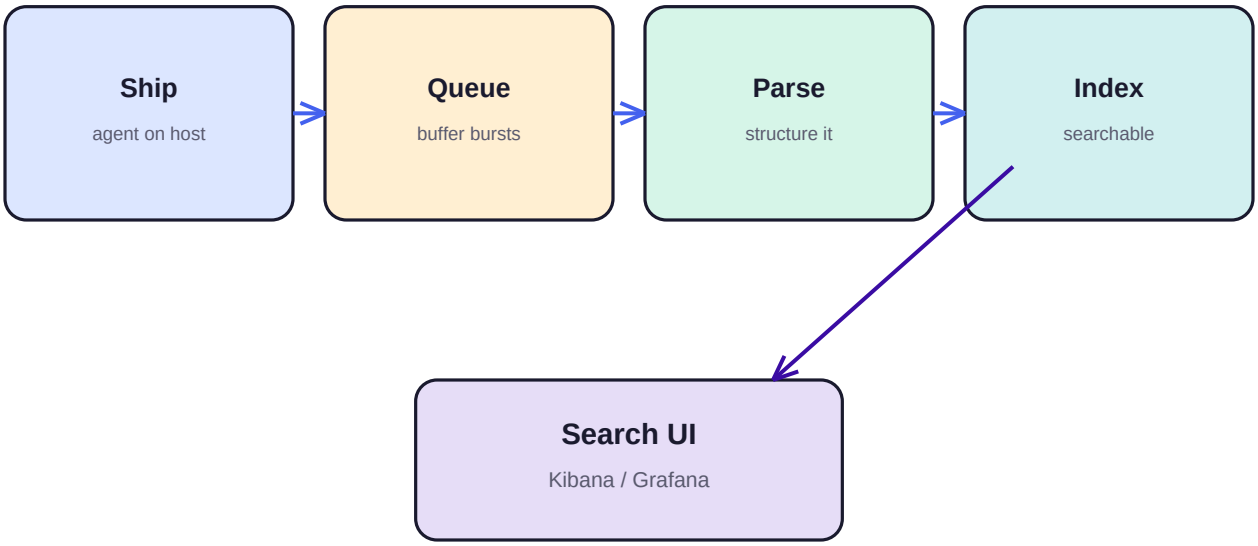
Ship, parse, index, search —
one pane of glass for every log line.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Log Pipeline

Four stages from app to searchable index.

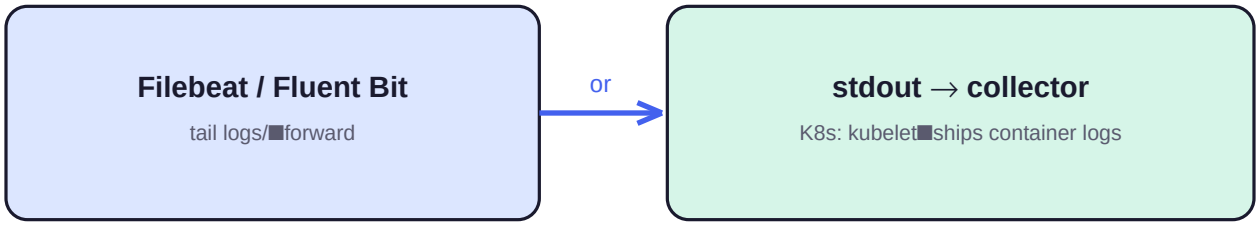


The queue (Kafka/Redis) absorbs spikes so a log storm never loses data or backpressures apps.

STEP 2/6

Shippers: Getting Logs Off the Box

Lightweight agents tail files and forward.



- Apps log to stdout/stderr — the platform collects, never local files in prod.
- Shippers are dumb pipes: minimal parsing at the edge, structure centrally.
- Backpressure handling: shipper buffers to disk when the pipeline is down.

STEP 3/6

ELK vs Loki: Index Everything vs Index Labels

Two philosophies of log storage

ELK
index full text

- Full-text search on every field
- Heavy: storage + compute hungry
- Great for ad-hoc exploration

Loki
index labels only

- Cheap: only label index, grep the rest
- Pairs with Prometheus/Grafana
- Needs good labels to be fast

Loki's mantra: 'like Prometheus, but for logs.' Same label model — cheap at scale, weaker full-text.

STEP 4/6

Structured Logging

JSON logs are queryable; printf logs are archaeology.

BAD: "user 42 failed login from 1.2.3.4"

GOOD: {"user_id":42,"event":"login_failed","ip":"1.2.3.4"}

Fields to always include: timestamp (UTC, ISO8601), level, service, trace_id, request_id.

- Log at the right level: DEBUG is dev-only; prod runs INFO+ (cost!).
- One line per event — multi-line stack traces need special handling to stay together.
- Never log secrets, tokens, or full PII — logs are broadly readable by design.

STEP 5/6

Retention & Tiers

Not all logs deserve SSDs forever.

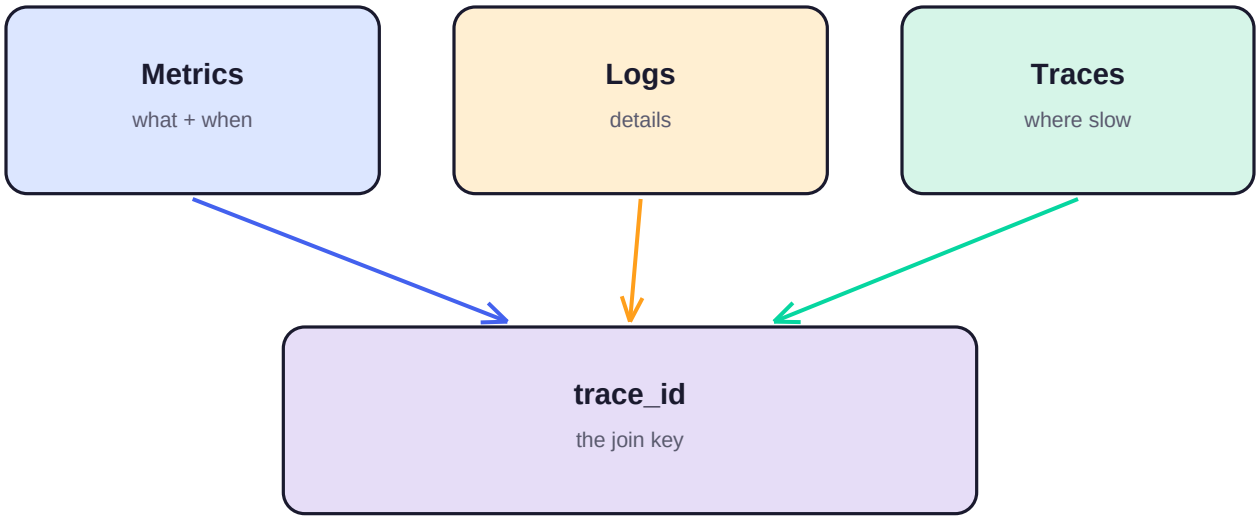
Index lifecycle policies move data down tiers automatically — set it once, forget it.

Log volume = money. Sample debug logs, drop health-check noise, and audit top talkers monthly.

STEP 6/6

Correlate: Logs + Metrics + Traces

A log line is 10x more useful with a trace_id in it.



Emit trace_id in every log line — jump from a slow trace straight to its logs in one click.

STEP ✓/✓

Key Takeaways

Centralized Logging — Remember

- Pipeline: ship → queue (buffer) → parse → index → search UI.
- Apps log to stdout; shippers forward; the queue absorbs storms.
- ELK indexes everything (powerful, pricey); Loki indexes labels (cheap, Prometheus-like).
- Structure logs as JSON with trace_id — printf logs are archaeology.
- Tier retention (hot/warm/cold) and audit volume — logs are money.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev